

Week 1 Lecture Prepare Quiz: Supervised Learning and Nearest Neighbours

- Due Sep 16 at 11a.m.
- Points 19
- Questions 19
- Available Sep 4 at 10a.m. - Sep 16 at 11a.m.
- Time Limit None
- Allowed Attempts Unlimited

Instructions

To prepare for this quiz, please read the following chapters from the course notes:

- [Introduction](https://www.teach.cs.toronto.edu/~csc311notes/intro.html)  (<https://www.teach.cs.toronto.edu/~csc311notes/intro.html>)
- Chapter [1.1](https://www.teach.cs.toronto.edu/~csc311notes/supervised_learning/supervised.html)  (https://www.teach.cs.toronto.edu/~csc311notes/supervised_learning/supervised.html) and [1.2](https://www.teach.cs.toronto.edu/~csc311notes/supervised_learning/knn.html)  (https://www.teach.cs.toronto.edu/~csc311notes/supervised_learning/knn.html)

Note that, unlike the other quizzes in the course, you will have **two weeks** to complete this quiz because it is the first quiz of the semester.

This quiz was locked Sep 16 at 11a.m..

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	13 minutes	19 out of 19

 Correct answers are hidden.

Score for this attempt: 19 out of 19

Submitted Sep 16 at 8:33a.m.

This attempt took 13 minutes.



Question 1

1 / 1 pts

Which of the following best describes supervised learning?

- ☐ Learning patterns from unlabeled inputs by grouping similar examples
- ☒ Learning to map inputs to outputs using labelled input-output pairs
- ☐ Memorizing the training set so that all future predictions are exact
- ☐ Choosing features without needing target outputs



Question 2

1 / 1 pts

Which of the following tasks is a classification task?

- ☐ Predicting tomorrow's highest temperature in Celsius
- ☐ Predicting a person's height from shoe size
- ☒ Predicting whether a leaf is oak or maple
- ☐ Predicting the age of a person in years from a social media message



Question 3

1 / 1 pts

What is the difference between t and y in the standard notation?

- ☐ t is the predicted output and y is the target output
- ☒ t is the target output and y is the predicted output
- ☐ t is a feature and y is a label
- ☐ t is used only for regression and y only for classification



Question 4

1 / 1 pts

Which of the following best matches the meaning of a hypothesis f ?

- ☒ A rule or mapping that takes an input x and produces a prediction
- ☐ A dataset of labelled examples used during training
- ☐ A method for evaluating generalization on unseen data
- ☐ A collection of models that all share the same hyperparameters



Question 5

1 / 1 pts

Which quantity determines whether a supervised learning problem is regression or classification?

- ☐ The type of input data
- ☐ The number of features
- ☒ The kind of output being predicted
- ☐ The size of the training set



Question 6

1 / 1 pts

Which dataset should be used to choose between different models?

- ☐ Training set
- ☒ Validation set
- ☐ Test set
- ☐ Entire labeled dataset combined



Question 7

1 / 1 pts

In the notation $\mathbf{x}^{(i)}$, what does the superscript (i) indicate?

- ☒ The i -th input vector in the dataset
- ☐ The i -th feature of a single example
- ☐ The i -th class label
- ☐ The exponent applied to the vector $\mathbf{x}$



Question 8

1 / 1 pts

Which of the following is an example of data augmentation for MNIST?

- ☐ Changing every label to the most common class
- ☐ Replacing grayscale values with color channels
- ☒ Shifting or rotating the image
- ☐ Removing difficult examples from the training set



Question 9

1 / 1 pts

Which of the following is the correct idea behind k-nearest neighbours?

- ☐ Predict by calculating the distance to the centroid of each class
- ☐ Predict using the average feature vector of the training set
- ☒ Predict using the majority class among the k closest training examples
- ☐ Partition the training data into k clusters and assign the query point to the nearest cluster center



Question 10

1 / 1 pts

Which of the following is said to be especially useful when the direction of vectors matters more than their magnitude, such as in text classification?

- ☐ Euclidean distance
- ☐ Manhattan distance
- ☐ Chebyshev distance
- ☒ Cosine similarity



Question 11

1 / 1 pts

The number k in k nearest neighbours is an example of what?

- ☐ Feature
- ☐ Label
- ☒ Hyperparameter
- ☐ Decision boundary



Question 12

1 / 1 pts

Compared with 1-NN, a 3-NN typically has a decision boundary that is:

- ☒ Smoother and less sensitive to individual training points
- ☐ Identical whenever Euclidean distance is used
- ☐ More jagged because three points create more boundaries
- ☐ Guaranteed to achieve higher training accuracy



Question 13

1 / 1 pts

Unless there are identical points with different labels, 1-NN achieves 100% training accuracy.

- ☒ True
- ☐ False



Question 14

1 / 1 pts

What happens in the extreme case $k=N$, where N is the size of the training set?

- ☐ The model predicts the label of the nearest point only
- ☒ The model always predicts the most common class in the training set
- ☐ The model becomes equivalent to 1-NN
- ☐ The model achieves zero training error by construction



Question 15

1 / 1 pts

In practice, how do we choose the value of k for a k nearest neighbour model for a given dataset?

- ☐ By always setting $k=1$
- ☐ By selecting the largest possible k
- ☒ By evaluating different values on a validation set
- ☐ By computing it directly from the test set



Question 16

1 / 1 pts

If feature A is measured on a much larger scale than feature B, then in k-NN:

- ☐ Differences in feature B have a larger effect on neighbor selection
- ☐ The algorithm automatically rescales all features before computing distances
- ☐ Both features always contribute equally to the distance calculation
- ☒ Differences in feature A can have a larger effect on distance calculations



Question 17

1 / 1 pts

A classifier has perfect accuracy on its training set. What is the most likely conclusion?

- ☐ It will necessarily achieve perfect test accuracy
- ☒ It may have overfit the training set
- ☐ It must be less complex than any classifier with lower training accuracy
- ☐ It should be trained on less data



Question 18

1 / 1 pts

In high dimensions, what happens to the Euclidean distances between randomly sampled points?

- ☐ Most points are clustered near the origin
- ☐ Most points are either very close or very far apart
- ☒ Most points are far apart from each other
- ☐ Distances decrease as dimensionality increases



Question 19

1 / 1 pts

Which of the following statements describe the curse of dimensionality in the context of k-NN? Select all that apply.

- ☒ k-NN may require much more data to make reliable predictions in high dimensions.
- ☐ Increasing the number of dimensions guarantees higher prediction accuracy
- ☒ In high dimensions, the nearest and farthest neighbors can have similar distances
- ☐ The curse of dimensionality disappears if k is chosen large enough.

Quiz Score: 19 out of 19